

## Java Assignment 1

//Q-1: Using javap, view the methods of the following classes from the lang package: java.lang.Object , java.lang.String and java.util.Scanner. and also Compile sample program 8. Type the following command and view the bytecodes. Javap -c MyClass.

```
Package Java.Assignment1;
```

```
Public class A1 {
```

```
    Public static void main(String[] args) {
```

```
        Int num = 7;
```

```
        Int square = num * num;
```

```
        System.out.println("Square = " + square);
```

```
    }
```

```
}
```

// Q-2:Write a program to calculate perimeter and area of rectangle. (hint : area = length \* breadth , perimeter=2\*(length+breadth))

```
package Assignment1;
```

```
import java.util.Scanner;
```

```
public class A2 {
```

```
    public static void main(String args[])
```

```
    {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter length: ");
```

```
        int length = sc.nextInt();
```

```
        System.out.print("Enter breadth: ");
```

```
        int breadth = sc.nextInt();
```

```
        float area = length*breadth;
```

```
        int perimeter = (2*(length*breadth));
```

```
        System.out.println("Area is : " +area);
```

```
        System.out.println("perimeter is : " +perimeter);
```

```
    }
```

```
}
```

// Q-3:Write a menu driven program to perform the following operations i.Calculate the volume of cylinder. (hint : Volume:  $\pi \times r^2 \times h$ ) ii.Find the factorial of given number. iii.Check the number is Armstrong or not. iv.Exit

```
Package Java.Assignment1;
```

```
Import java.util.Scanner;
```

```
Public class A3 {
```

```
    Public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        Int ch;
```

```
        Do {
```

```
            System.out.println("\n1.Volume of Cylinder\n2.Factorial\n3.Armstrong\n4.Exit");
```

```
            System.out.print("Enter your choice:"); // ask user for choice
```

```
            Ch = sc.nextInt();
```

```
            Switch (ch) {
```

```
                Case 1:
```

```
                    System.out.print("Enter radius:");
```

```
                    Double r = sc.nextDouble();
```

```
                    System.out.print("Enter height:"); // ask user for height of cylinder
```

```
                    Double h = sc.nextDouble();
```

```
                    System.out.println("Volume of cylinder=" + (Math.PI * r * r * h));
```

```
                    Break;
```

```
                Case 2:
```

```
                    System.out.print("Enter number:");
```

```
                    Int n = sc.nextInt(), f = 1;
```

```
                    For (int i = 1; i <= n; i++) f *= i;
```

```
                    System.out.println("Factorial of " + n + "=" + f);
```

```
                    Break;
```

```
                Case 3:
```

```

        System.out.print("Enter number:");

        Int num = sc.nextInt(), sum = 0, t = num, d, digits = 0;

        For (int x = num; x > 0; x /= 10) digits++;

        For (; t > 0; t /= 10) { d = t % 10; sum += Math.pow(d, digits); }

        System.out.println(sum == num ? "Armstrong" : "Not Armstrong");

        Break;

        Default: System.out.println("Bye");

    }

} while (ch != 4);

}

}

```

// Q-4: Write a program to accept the array element and display in reverse order.

```

package Java.Assignment1;

import java.util.*;

public class A4 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter size of array: ");

        int n = sc.nextInt();

        int[] arr = new int[n];

        System.out.println("Enter " + n + " elements:");

        for (int i = 0; i < n; i++) {

            arr[i] = sc.nextInt();

        }

        System.out.println("Array in reverse order:");

        for (int i = n - 1; i >= 0; i--) {

            System.out.print(arr[i] + " ");

        }

    }

}

```

```
}  
}
```

// Q-5:Write a java program to display the system date and time in various formats shown below:

```
// Current date is : 31/08/2021
```

```
// Current date is : 08-31-2021
```

```
// Current date is : Tuesday August 31 2021
```

```
// Current date and time is : Fri August 31 15:25:59 IST 2021
```

```
// Current date and time is : 31/08/21 15:25:59 PM +0530
```

```
// Current time is : 15:25:59
```

```
// Current week of year is : 35
```

```
// Current week of month : 5
```

```
// Current day of the year is : 243
```

```
// Note: Use java.util.Date and java.text.SimpleDateFormat class
```

```
Package Java.Assignment1;
```

```
Import java.util.Date;
```

```
Import java.util.Scanner;
```

```
Import java.text.SimpleDateFormat;
```

```
Import java.util.Calendar;
```

```
Public class B1{
```

```
    Public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter date (dd/MM/yyyy): ");
```

```
        String input = sc.nextLine();
```

```
        Date now = new Date();
```

```
        Calendar cal = Calendar.getInstance();
```

```
        System.out.println("Current date is : " + new  
SimpleDateFormat("dd/MM/yyyy").format(now));
```

```

        System.out.println("Current date is : " + new SimpleDateFormat("MM-dd-
yyyy").format(now));

        System.out.println("Current date is : " + new SimpleDateFormat("EEEE MMMM dd
yyyy").format(now));

        System.out.println("Current date and time is : " + new SimpleDateFormat("E
MMMM dd HH:mm:ss z yyyy").format(now));

        System.out.println("Current date and time is : " + new
SimpleDateFormat("dd/MM/yy HH:mm:ss a Z").format(now));

        System.out.println("Current time is : " + new
SimpleDateFormat("HH:mm:ss").format(now));

        System.out.println("Current week of year is : " + cal.get(Calendar.WEEK_OF_YEAR));

        System.out.println("Current week of month : " +
cal.get(Calendar.WEEK_OF_MONTH));

        System.out.println("Current day of the year is : " + cal.get(Calendar.DAY_OF_YEAR));
    }
}

```

// Q-6:Write a menu driven program to perform the following operations on multidimensional array ie matrix :

// i.Addition

// ii.Multiplication

// iii.Transpose of any matrix.

// iv.Exit

Package Java.Assignment1;

Import java.util.Scanner;

Public class B3 {

Public static void main(String[] args) {

Scanner sc = new Scanner(System.in);

While (true) {

System.out.println("\n1. Add\n2. Multiply\n3. Transpose\n4. Exit");

System.out.print("Choose: ");

```

Int ch = sc.nextInt();

If (ch == 4) break;

System.out.print("Rows: ");

Int r = sc.nextInt();

System.out.print("Cols: ");

Int c = sc.nextInt();

Int[][] A = new int[r][c];

System.out.println("Enter Matrix:");

For (int i = 0; i < r; i++)

    For (int j = 0; j < c; j++)

        A[i][j] = sc.nextInt();


If (ch == 1) {

    Int[][] B = new int[r][c];

    System.out.println("Enter Second Matrix:");

    For (int i = 0; i < r; i++)

        For (int j = 0; j < c; j++)

            B[i][j] = sc.nextInt();

    System.out.println("Sum:");

    For (int i = 0; i < r; i++) {

        For (int j = 0; j < c; j++)

            System.out.print((A[i][j] + B[i][j]) + " ");

        System.out.println();

    }

} else if (ch == 2) {

    System.out.print("Cols of second matrix: ");

    Int c2 = sc.nextInt();

    Int[][] B = new int[c][c2];

```

```

        System.out.println("Enter Second Matrix:");

        For (int i = 0; i < c; i++)

            For (int j = 0; j < c2; j++)

                B[i][j] = sc.nextInt();

        Int[][] res = new int[r][c2];

        For (int i = 0; i < r; i++)

            For (int j = 0; j < c2; j++)

                For (int k = 0; k < c; k++)

                    Res[i][j] += A[i][k] * B[k][j];

        System.out.println("Product:");

        For (int[] row : res) {

            For (int val : row)

                System.out.print(val + " ");

            System.out.println();

        }

    } else if (ch == 3) {

        System.out.println("Transpose:");

        For (int i = 0; i < c; i++) {

            For (int j = 0; j < r; j++)

                System.out.print(A[j][i] + " ");

            System.out.println();

        }

    } else {

        System.out.println("Invalid choice.");

    }

}

Sc.close();

}

```

```
}
```

// Q-7:Write a program to accept n names of country and display them in descending order.

```
Package Java.Assignment1;
```

```
Import java.util.*;
```

```
Public class C1 {
```

```
    Public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter number of countries: ");
```

```
        Int n = sc.nextInt();
```

```
        Sc.nextLine(); // consume newlin
```

```
        String[] countries = new String[n];
```

```
        System.out.println("Enter country names:");
```

```
        For (int i = 0; i < n; i++) {
```

```
            Countries[i] = sc.nextLine();
```

```
        }
```

```
        Arrays.sort(countries, Collections.reverseOrder());
```

```
        System.out.println("\nCountries in descending order:");
```

```
        For (String country : countries) {
```

```
            System.out.println(country);
```

```
        }
```

```
    }
```

```
}
```

// Q-8:Write a menu driven program to perform the following operations on 2D array:

// i.Sum of diagonal elements

// ii.Sum of upper diagonal elements

// iii.Sum of lower diagonal elements

// iv.Exit



```
package Java.Assignment1;

import java.util.Scanner;

public class C2 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter size of square matrix: ");

        int n = sc.nextInt();

        int[][] mat = new int[n][n];

        System.out.println("Enter matrix elements:");

        for (int i = 0; i < n; i++)

            for (int j = 0; j < n; j++)

                mat[i][j] = sc.nextInt();

        int choice;

        do {

            System.out.println("\n1. Sum of Diagonal");

            System.out.println("2. Sum of Upper Diagonal");

            System.out.println("3. Sum of Lower Diagonal");

            System.out.println("4. Exit");

            System.out.print("Enter your choice: ");

            choice = sc.nextInt();

            int sum = 0;

            switch (choice) {

                case 1:

                    for (int i = 0; i < n; i++)

                        sum += mat[i][i];

                    System.out.println("Sum of diagonal elements: " + sum);

                    break;

                case 2:
```

```

        for (int i = 0; i < n; i++)
            for (int j = i + 1; j < n; j++)
                sum += mat[i][j];
        System.out.println("Sum of upper diagonal elements: " + sum);
        break;
    case 3:
        for (int i = 0; i < n; i++)
            for (int j = 0; j < i; j++)
                sum += mat[i][j];
        System.out.println("Sum of lower diagonal elements: " + sum);
        break;
    case 4:
        System.out.println("Exiting program.");
        break;
    default:
        System.out.println("Invalid choice.");
    }
} while (choice != 4);
sc.close();
}
}

```

// Q-9:Write a program to display the 1 to 15 tables.

```
Package Java.Assignment1;
```

```
Public class C3 {
```

```
    Public static void main(String[] args) {
```

```
        For (int i = 1; i <= 10; i++) {
```

```
            For (int j = 1; j <= 15; j++) {
```

```
                System.out.printf("%-9s", (j + "*" + i + "=" + (i * j)));
```

```

    }

    System.out.println();

}

System.out.println("Multiplication table from 0 to 10 and 0 to 15 is printed.");

}

}

```

## Java Assignment2

// Q-1: Define Student class(roll\_no, name, percentage) to create n objects of the Student class. Accept details from the user for each object. Define a static method "sortStudent" which sorts the array on the basis of percentage.

```

// import java.util.*;

Class Student {
    Int roll_no;
    String name;
    Double percentage;
    Student(int roll_no, String name, double percentage) {
        This.roll_no = roll_no;
        This.name = name;
        This.percentage = percentage;
    }
    Public static void sortStudent(Student[] students) {
        Arrays.sort(students, Comparator.comparingDouble(s -> s.percentage));
    }
    Public void display() {
        System.out.println("Roll No: " + roll_no + ", Name: " + name + ", Percentage: " +
percentage);
    }
}

Public class A2 {

```

```

Public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter number of students: ");

    Int n = sc.nextInt();

    Student[] students = new Student[n];

    For (int i = 0; i < n; i++) {

        System.out.println("\nEnter details for student " + (i + 1) + " :");

        System.out.print("Roll No: ");

        Int roll_no = sc.nextInt();

        Sc.nextLine(); // consume newline

        System.out.print("Name: ");

        String name = sc.nextLine();

        System.out.print("Percentage: ");

        Double percentage = sc.nextDouble();

        Students[i] = new Student(roll_no, name, percentage);

    }

    Student.sortStudent(students);

    System.out.println("\nStudents sorted by percentage:");

    For (Student s : students) {

        s.display();

    }

    Sc.close();

}

}

// Q-2:Write a java program to accept 5 numbers using command line arguments sort
and display them.

Import java.util.Arrays;

Public class A3 {

```

```

    public static void main(String[] args) {
        if (args.length != 5) {
            System.out.println("Please enter exactly 5 numbers as command-line arguments.");
            return;
        }
        int[] numbers = new int[5];
        for (int i = 0; i < 5; i++) {
            try {
                numbers[i] = Integer.parseInt(args[i]);
            } catch (NumberFormatException e) {
                System.out.println("Invalid input: " + args[i] + " is not a valid number.");
                return;
            }
        }
        Arrays.sort(numbers);
        System.out.println("Sorted numbers:");
        for (int num : numbers) {
            System.out.print(num + " ");
        }
    }
}

```

// Q-3: Write a java program that take input as a person name in the format of first, middle and last name and then print it in the form last, first and middle name, where in the middle name first character is capital letter.

```

import java.util.Scanner;

public class A4 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
    }
}

```

```

        System.out.print("Enter full name (first middle last): ");

        String first = sc.next();

        String middle = sc.next();

        String last = sc.next();

        String middleFormatted = middle.substring(0, 1).toUpperCase() +
        middle.substring(1);

        System.out.println("Formatted name: " + last + ", " + first + " " + middleFormatted);

        sc.close();

    }

}

```

// Q-4: Write a Java program to create a Package "SY" which has a class SYMarks (members – ComputerTotal, MathsTotal, and ElectronicsTotal). Create another package TY which has a class TYMarks (members – Theory, Practicals). Create n objects of Student class (having rollNumber, name, SYMarks and TYMarks). Add the marks of SY and TY computer subjects and calculate the Grade ('A' for >= 70, 'B' for >= 60 'C' for >= 50, Pass Class for >= 40 else 'FAIL') and display the result of the student in proper format.

```

Import java.util.Scanner;

```

```

Class SYMarks {

```

```

    Int computerTotal, mathsTotal, electronicsTotal;

```

```

    SYMarks(int c, int m, int e) {

```

```

        computerTotal = c;

```

```

        mathsTotal = m;

```

```

        electronicsTotal = e;

```

```

    }

```

```

}

```

```

Class TYMarks {

```

```

    Int theory, practicals;

```

```

    TYMarks(int t, int p) {

```

```

        Theory = t;

```

```

        Practicals = p;
    }
}

Class Student {
    Int rollNumber;
    String name;
    SYMarks sy;
    TYMarks ty;
    Student(int roll, String nm, SYMarks syMarks, TYMarks tyMarks) {
        rollNumber = roll;
        name = nm;
        sy = syMarks;
        ty = tyMarks;
    }
    Void displayResult() {
        Int totalComputerMarks = sy.computerTotal + ty.theory + ty.practicals;
        String grade;
        If (totalComputerMarks >= 70) grade = "A";
        Else if (totalComputerMarks >= 60) grade = "B";
        Else if (totalComputerMarks >= 50) grade = "C";
        Else if (totalComputerMarks >= 40) grade = "Pass Class";
        Else grade = "FAIL";
        System.out.println("\nRoll No: " + rollNumber);
        System.out.println("Name : " + name);
        System.out.println("SY Computer Marks : " + sy.computerTotal);
        System.out.println("TY Theory : " + ty.theory);
        System.out.println("TY Practicals : " + ty.practicals);
        System.out.println("Total Computer Marks: " + totalComputerMarks);
    }
}

```

```

        System.out.println("Grade: " + grade);
    }
}

Public class B1 {

    Public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of students: ");

        Int n = sc.nextInt();

        Student[] students = new Student[n];

        For (int i = 0; i < n; i++) {

            System.out.println("\nEnter details for student " + (i + 1));

            System.out.print("Roll Number: ");

            Int roll = sc.nextInt();

            Sc.nextLine();

            System.out.print("Name: ");

            String name = sc.nextLine();

            System.out.print("SY Computer Marks: ");

            Int syComp = sc.nextInt();

            System.out.print("SY Maths Marks: ");

            Int syMaths = sc.nextInt();

            System.out.print("SY Electronics Marks: ");

            Int syElect = sc.nextInt();

            System.out.print("TY Theory Marks: ");

            Int tyTheory = sc.nextInt();

            System.out.print("TY Practicals Marks: ");

            Int tyPract = sc.nextInt();

            SYMarks sy = new SYMarks(syComp, syMaths, syElect);

            TYMarks ty = new TYMarks(tyTheory, tyPract);

```



```

        Students[i] = new Student(roll, name, sy, ty);
    }

    System.out.println("\n--- Student Results ---");

    For (Student s : students) {
        s.displayResult();
    }
}

```

// Q-5: Define a class CricketPlayer (name, no\_of\_innings, no\_of\_times\_notout, totatruns, bat\_avg). Create an array of n player objects. Calculate the batting average for each player using static method avg(). Define a static sort method which sorts the array on the basis of average. Display the player details in sorted order.

```

// import java.util.Scanner;

class CricketPlayer {
    String name;
    int no_of_innings;
    int no_of_times_notout;
    int total_runs;
    float bat_avg;

    CricketPlayer(String name, int innings, int notout, int runs) {
        this.name = name;
        this.no_of_innings = innings;
        this.no_of_times_notout = notout;
        this.total_runs = runs;
        this.bat_avg = avg(innings, notout, runs);
    }

    static float avg(int innings, int notout, int runs) {
        int dismissals = innings - notout;
        if (dismissals == 0) return runs;
    }
}

```

```

        return (float) runs / dismissals;
    }

    static void sort(CricketPlayer[] players) {
        for (int i = 0; i < players.length - 1; i++) {
            for (int j = i + 1; j < players.length; j++) {
                if (players[j].bat_avg > players[i].bat_avg) {
                    CricketPlayer temp = players[i];
                    players[i] = players[j];
                    players[j] = temp;
                }
            }
        }
    }

    void display() {
        System.out.printf("%-15s %-10d %-10d %-10d %-10.2f\n",
            name, no_of_innings, no_of_times_notout, total_runs, bat_avg);
    }
}

public class B2 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of players: ");

        int n = sc.nextInt();

        CricketPlayer[] players = new CricketPlayer[n];

        for (int i = 0; i < n; i++) {
            System.out.println("\nEnter details for player " + (i + 1));

            sc.nextLine(); // clear buffer

            System.out.print("Name: ");

```

```

        String name = sc.nextLine();

        System.out.print("No. of Innings: ");

        int innings = sc.nextInt();

        System.out.print("No. of Times Not Out: ");

        int notout = sc.nextInt();

        System.out.print("Total Runs: ");

        int runs = sc.nextInt();

        players[i] = new CricketPlayer(name, innings, notout, runs);
    }

    CricketPlayer.sort(players);

    System.out.println("\n--- Player Details Sorted by Batting Average ---");

    System.out.printf("%-15s %-10s %-10s %-10s %-10s\n",
        "Name", "Innings", "NotOut", "Runs", "Average");

    for (CricketPlayer p : players) {
        p.display();
    }
}

```

// Q-6:Write a package for String operation which has two classes Con and Comp. Con class has to concatenates two strings and comp class compares two strings. Also display proper message on execution.

```

import java.util.Scanner;

class Con {

    void concatenate(String s1, String s2) {

        String result = s1 + s2;

        System.out.println("Concatenated String: " + result);

    }

}

class Comp {

```

```

void compare(String s1, String s2) {
    if (s1.equals(s2)) {
        System.out.println("Strings are equal.");
    } else {
        System.out.println("Strings are not equal.");
    }
}
}

public class C1 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter first string: ");
        String str1 = sc.nextLine();
        System.out.print("Enter second string: ");
        String str2 = sc.nextLine();
        Con conObj = new Con();
        Comp compObj = new Comp();
        System.out.println("\n--- String Operations ---");
        conObj.concatenate(str1, str2);
        compObj.compare(str1, str2);
    }
}

```

// Q-7: Create four member variables for Customer class. Assign public, private, protected and default access modifiers respectively to these variables. Try to access these variables from other classes (Same package and Different package)

```

Class Customer {
    Public String publicName = "Public Name";
    Private String privateEmail = private@email.com;
    Protected String protectedPhone = "9876543210";
}

```

```

String defaultCity = "Pune";

Void showFromCustomer() {
    System.out.println("Inside Customer class:");
    System.out.println("Public: " + publicName);
    System.out.println("Private: " + privateEmail);
    System.out.println("Protected: " + protectedPhone);
    System.out.println("Default: " + defaultCity);
}
}

Class SamePackageAccess {
    Void showFromSamePackage() {
        Customer c = new Customer();
        System.out.println("\nAccess from SamePackageAccess class:");
        System.out.println("Public: " + c.publicName);
        System.out.println("Protected: " + c.protectedPhone);
        System.out.println("Default: " + c.defaultCity);
    }
}

Class DifferentPackageAccess extends Customer {
    Void showFromDifferentPackage() {
        System.out.println("\nAccess from DifferentPackageAccess class (simulating
different package):");
        System.out.println("Public: " + publicName);
        System.out.println("Protected (via inheritance): " + protectedPhone);
    }
}

Public class C2 {
    Public static void main(String[] args) {

```

```

    Customer c = new Customer();

    c.showFromCustomer();

    SamePackageAccess spa = new SamePackageAccess();

    Spa.showFromSamePackage();

    DifferentPackageAccess dpa = new DifferentPackageAccess();

    Dpa.showFromDifferentPackage();

}

}

```

### Java Assignment 3

// Q-1: Define an abstract class Staff with protected members id and name. Define a parameterized constructor. Define one subclass OfficeStaff with member department. Create n objects of OfficeStaff and display all details.

```

// import java.util.Scanner;

// Abstract class Staff

abstract class Staff {

    protected int id;

    protected String name;

    public Staff(int id, String name) {

        this.id = id;

        this.name = name;

    }

    // Abstract method to display details

    public abstract void display();

}

// Subclass OfficeStaff

class OfficeStaff extends Staff {

    private String department;

    public OfficeStaff(int id, String name, String department) {

        super(id, name);

    }

}

```

```

        this.department = department;
    }

    // Override display method
    public void display() {
        System.out.println("ID: " + id);
        System.out.println("Name: " + name);
        System.out.println("Department: " + department);
        System.out.println("-----");
    }
}

// Main class
public class a2 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of OfficeStaff: ");
        int n = sc.nextInt();
        sc.nextLine(); // consume newline
        OfficeStaff[] staffArr = new OfficeStaff[n];
        for (int i = 0; i < n; i++) {
            System.out.print("Enter ID: ");
            int id = sc.nextInt();
            sc.nextLine();
            System.out.print("Enter Name: ");
            String name = sc.nextLine();
            System.out.print("Enter Department: ");
            String dept = sc.nextLine();
            staffArr[i] = new OfficeStaff(id, name, dept);
        }
    }
}

```

```

        System.out.println("All OfficeStaff Details:");
        for (OfficeStaff staff : staffArr) {
            staff.display();
        }
        sc.close();
    }
}

```

// Q-2: Define an interface "Operation" which has methods area(), volume(). Define a constant PI having a value 3.142. Create a class cylinder which implements this interface (members – radius, height) Create one object and calculate the area and volume.

```

Interface Operation {
    Double PI = 3.142; // Constant defined in interface
    Double area(); // Abstract method
    Double volume(); // Abstract method
}

Class Cylinder implements Operation {
    Private double radius;
    Private double height;
    Public Cylinder(double radius, double height) {
        This.radius = radius;
        This.height = height;
    }
    // Area of Cylinder:  $2 * PI * r * h + 2 * PI * r * r$ 
    Public double area() {
        Return (2 * PI * radius * height) + (2 * PI * radius * radius);
    }
    // Volume of Cylinder:  $PI * r^2 * h$ 
    Public double volume() {

```



```

        Return PI * radius * radius * height;
    }

    // Display results
    Public void display() {
        System.out.printf("Area of Cylinder: %.3f", area());
        System.out.printf("Volume of Cylinder: %.3f", volume());
    }
}

Public class a3 {
    Public static void main(String[] args) {
        Cylinder c = new Cylinder(5, 10); // Example: radius = 5, height = 10
        c.display();
    }
}

// Q-3:Write a program to find the cube of given number using function interface. Define
a functional interface

@FunctionalInterface
Interface Cube {
    Int calculate(int number);
}

Public class a4 {
    Public static void main(String[] args) {
        // Lambda expression implementing the Cube interface
        Cube cube = (number) -> number * number * number;
        Int num = 5;
        Int result = cube.calculate(num);
        System.out.println("Cube of " + num + " is: " + result);
    }
}

```

```
}
```

// Q-4:Create an abstract class “order” having members id,description.Create two subclasses “Purchase Order” and “Sales Order” having members customer name and Vendor name respectively.Define methods accept and display in all cases. Create 3 objects each of Purchase Order and Sales Order and accept and display details.

```
Import java.util.Scanner;
```

```
// Abstract class Order
```

```
Abstract class Order {
```

```
    Int id;
```

```
    String description;
```

```
    Abstract void accept(Scanner sc);
```

```
    Abstract void display();
```

```
}
```

```
// Subclass PurchaseOrder
```

```
Class PurchaseOrder extends Order {
```

```
    String customerName;
```

```
    Void accept(Scanner sc) {
```

```
        System.out.print("Enter Purchase Order ID: ");
```

```
        Id = sc.nextInt();
```

```
        Sc.nextLine(); // consume newline
```

```
        System.out.print("Enter Purchase Order Description: ");
```

```
        Description = sc.nextLine();
```

```
        System.out.print("Enter Customer Name: ");
```

```
        customerName = sc.nextLine();
```

```
    }
```

```
    Void display() {
```

```
        System.out.println("Purchase Order ID: " + id);
```

```
        System.out.println("Description: " + description);
```

```
        System.out.println("Customer Name: " + customerName);
```

```

        System.out.println();
    }
}

// Subclass SalesOrder
Class SalesOrder extends Order {
    String vendorName;

    Void accept(Scanner sc) {
        System.out.print("Enter Sales Order ID: ");
        Id = sc.nextInt();

        Sc.nextLine(); // consume newline

        System.out.print("Enter Sales Order Description: ");
        Description = sc.nextLine();

        System.out.print("Enter Vendor Name: ");
        vendorName = sc.nextLine();
    }

    Void display() {
        System.out.println("Sales Order ID: " + id);
        System.out.println("Description: " + description);
        System.out.println("Vendor Name: " + vendorName);
        System.out.println();
    }
}

// Main class to test the functionality
Public class b1 {
    Public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        PurchaseOrder[] purchaseOrders = new PurchaseOrder[3];
        SalesOrder[] salesOrders = new SalesOrder[3];
    }
}

```

```

System.out.println("Enter details for 3 Purchase Orders:");
For (int i = 0; i < 3; i++) {
    purchaseOrders[i] = new PurchaseOrder();
    purchaseOrders[i].accept(sc);
}
System.out.println("Enter details for 3 Sales Orders:");
For (int i = 0; i < 3; i++) {
    salesOrders[i] = new SalesOrder();
    salesOrders[i].accept(sc);
}
System.out.println("Purchase Orders Details:");
For (PurchaseOrder po : purchaseOrders) {
    Po.display();
}
System.out.println("Sales Orders Details:");
For (SalesOrder so : salesOrders) {
    So.display();
}
Sc.close();
}
}

// Q-5:Write a program to using marker interface create a class
product(product_id,product_name, product_cost, product_quantity) define a default
and parameterized constructor. Create objects of class product and display the
contents of each object and Also display the object count.

// Marker interface
Interface ProductMarker {
    // empty marker interface
}

```

// Product class implementing marker interface

Class Product implements ProductMarker {

Int product\_id;

String product\_name;

Double product\_cost;

Int product\_quantity;

Static int objectCount = 0; // static count of objects

// Default constructor

Public Product() {

    This.product\_id = 0;

    This.product\_name = "Default";

    This.product\_cost = 0.0;

    This.product\_quantity = 0;

    objectCount++;

}

// Parameterized constructor

Public Product(int product\_id, String product\_name, double product\_cost, int product\_quantity) {

    This.product\_id = product\_id;

    This.product\_name = product\_name;

    This.product\_cost = product\_cost;

    This.product\_quantity = product\_quantity;

    objectCount++;

}

// Method to display product details

Public void display() {

    System.out.println("Product ID: " + product\_id);

    System.out.println("Product Name: " + product\_name);

```

        System.out.println("Product Cost: " + product_cost);
        System.out.println("Product Quantity: " + product_quantity);
        System.out.println("-----");
    }
    // Static method to display object count
    Public static void displayObjectCount() {
        System.out.println("Total Product Objects Created: " + objectCount);
    }
}

```

```

Public class b2 {
    Public static void main(String[] args) {
        // Creating product objects
        Product p1 = new Product(); // default constructor
        Product p2 = new Product(101, "Laptop", 75000.50, 5);
        Product p3 = new Product(102, "Smartphone", 32000.00, 10);
        // Display product details
        P1.display();
        P2.display();
        P3.display();
        // Display total object count
        Product.displayObjectCount();
    }
}

```

// Q-6: Create an interface Department containing attributes deptName and deptHead. It also has abstract methods for printing the attributes. Create a class hostel containing hostelName, hostelLocation and numberOfRooms. The class contains method printing the attributes. Then write Student class extending the Hostel class and implementing the Department interface. This class contains attributes studentName, regNo, electiveSubject and avgMarks. Write suitable printData method for this class. Also,

implement the abstract methods of the Department interface. Write a driver class to test the Student class. The program should be menu driven containing the options:

// i.Admit new student

// ii.Migrate a student

// iii.Display details of a student

// For the third option, a search is to be made on the basis of the entered registration Number.

Import java.util.ArrayList;

Import java.util.Scanner;

// Department Interface

Interface Department {

String deptName = null;

String deptHead = null;

Void printDeptName();

Void printDeptHead();

}

// Hostel Class

Class Hostel {

String hostelName;

String hostelLocation;

Int numberOfRooms;

Public Hostel() {}

Public Hostel(String hostelName, String hostelLocation, int numberOfRooms) {

This.hostelName = hostelName;

This.hostelLocation = hostelLocation;

This.numberOfRooms = numberOfRooms;

}

Void printHostelInfo() {

System.out.println("Hostel Name: " + hostelName);

```

        System.out.println("Hostel Location: " + hostelLocation);

        System.out.println("Number Of Rooms: " + numberOfRooms);
    }
}

// Student Class

Class Student extends Hostel implements Department {

    String studentName;

    String regNo;

    String electiveSubject;

    Double avgMarks;

    String deptName; // implement deptName from Department interface

    String deptHead; // implement deptHead from Department interface

    Public Student() {}

    Public Student(String hostelName, String hostelLocation, int numberOfRooms,

        String studentName, String regNo, String electiveSubject, double avgMarks,

        String deptName, String deptHead) {

        Super(hostelName, hostelLocation, numberOfRooms);

        This.studentName = studentName;

        This.regNo = regNo;

        This.electiveSubject = electiveSubject;

        This.avgMarks = avgMarks;

        This.deptName = deptName;

        This.deptHead = deptHead;

    }

    // Implement abstract methods of Department interface

    Public void printDeptName() {

        System.out.println("Department Name: " + deptName);

    }
}

```



```

Public void printDeptHead() {
    System.out.println("Department Head: " + deptHead);
}

// printData method
Void printData() {
    System.out.println("--- Student Details ---");
    System.out.println("Student Name: " + studentName);
    System.out.println("Registration Number: " + regNo);
    System.out.println("Elective Subject: " + electiveSubject);
    System.out.println("Average Marks: " + avgMarks);
    printDeptName();
    printDeptHead();
    printHostelInfo();
}
}

// Driver class
Public class c1 {
    Static ArrayList<Student> students = new ArrayList<>();
    Static Scanner scanner = new Scanner(System.in);
    // Method to admit new student
    Static void admitNewStudent() {
        System.out.print("Enter Hostel Name: ");
        String hName = scanner.nextLine();
        System.out.print("Enter Hostel Location: ");
        String hLocation = scanner.nextLine();
        System.out.print("Enter Number Of Rooms in Hostel: ");
        Int noRooms = Integer.parseInt(scanner.nextLine());
        System.out.print("Enter Student Name: ");
    }
}

```

```

String sName = scanner.nextLine();

System.out.print("Enter Registration Number: ");

String regNo = scanner.nextLine();

System.out.print("Enter Elective Subject: ");

String elective = scanner.nextLine();

System.out.print("Enter Average Marks: ");

Double avgMarks = Double.parseDouble(scanner.nextLine());

System.out.print("Enter Department Name: ");

String dName = scanner.nextLine();

System.out.print("Enter Department Head: ");

String dHead = scanner.nextLine();

Student newStudent = new Student(hName, hLocation, noRooms, sName, regNo,
elective, avgMarks, dName, dHead);

Students.add(newStudent);

System.out.println("Student admitted successfully.");
}

// Method to migrate a student (update details)

Static void migrateStudent() {

    System.out.print("Enter Registration Number of student to migrate: ");

    String regNo = scanner.nextLine();

    Student foundStudent = null;

    For (Student s : students) {

        If (s.regNo.equals(regNo)) {

            foundStudent = s;

            break;

        }

    }

    If (foundStudent == null) {

```

```

        System.out.println("Student with Registration Number " + regNo + " not found.");
        Return;
    }
    System.out.println("Enter updated details for migration:");
    System.out.print("Enter New Hostel Name: ");
    foundStudent.hostelName = scanner.nextLine();
    System.out.print("Enter New Hostel Location: ");
    foundStudent.hostelLocation = scanner.nextLine();
    System.out.print("Enter New Number Of Rooms: ");
    foundStudent.numberOfRooms = Integer.parseInt(scanner.nextLine());
    System.out.print("Enter New Department Name: ");
    foundStudent.deptName = scanner.nextLine();
    System.out.print("Enter New Department Head: ");
    foundStudent.deptHead = scanner.nextLine();
    System.out.println("Student migration details updated successfully.");
}

// Method to display student details by regNo
Static void displayStudent() {
    System.out.print("Enter Registration Number of student to display: ");
    String regNo = scanner.nextLine();
    Student foundStudent = null;
    For (Student s : students) {
        If (s.regNo.equals(regNo)) {
            foundStudent = s;
            break;
        }
    }
}

If (foundStudent == null) {

```

```

        System.out.println("Student with Registration Number " + regNo + " not found.");

        Return;
    }

    foundStudent.printData();
}

Public static void main(String[] args) {
    While (true) {
        System.out.println("Menu:");

        System.out.println("1. Admit new student");

        System.out.println("2. Migrate a student");

        System.out.println("3. Display details of a student");

        System.out.println("4. Exit");

        System.out.print("Enter your choice: ");

        Int choice = Integer.parseInt(scanner.nextLine());

        Switch (choice) {
            Case 1:
                admitNewStudent();

                break;
            case 2:
                migrateStudent();

                break;
            case 3:
                displayStudent();

                break;
            case 4:
                System.out.println("Exiting program.");

                System.exit(0);

            Default:

```

```
System.out.println("Invalid choice! Please enter again.");
```

```
}
```

```
}
```

```
}
```

```
}
```